

Total No. of Questions : 8]

SEAT No. :

PC-1782

[Total No. of Pages :2

[6353]-101

T.E. (E & TC)

DIGITAL IMAGE PROCESSING

(2019 Pattern) (Semester - II) (Elective - II) (304195 (A))

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Your answers will be valued as a whole.
- 5) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.
- 6) Assume suitable data, if necessary.

- Q1)** a) What is edge detection? Compare the performance of first order & second order derivative w.r.t. image. [6]
- b) Explain the Laplacian edge detector. Explain Why the LoG mask is preferred over the Laplacian edge detector. [6]
- c) What is Image thresholding? Explain Local, global, and adaptive thresholding. [6]

OR

- Q2)** a) Explain Image segmentation using. [6]
- i) Region growing.
 - ii) Region Splitting.
- b) With the help of suitable masks, explain the following. [6]
- i) Point detection.
 - ii) Line detection.
- c) Explain the following edge-detecting operators in detail. [6]
- i) Prewitt operator
 - ii) Sobel operator

- Q3)** a) What is the need of image compression? Explain an image compression model. [6]
- b) What is data redundancy? Explain the redundancies used in image compression. [5]
- c) Explain the need for fidelity criteria in image compression. Write any two fidelity measures. [6]

P.T.O.

OR

- Q4)** a) What is lossy compression technique? Explain the DCT-based compression technique. [6]
b) Explain the concept of motion estimation with the help of any one algorithm in detail. [6]
c) Draw and explain JPEG base line encoder. Comment on block size used in JPEG. [5]
- Q5)** a) Explain image restoration process with help of block diagram. [6]
b) Explain any three noise models in short. [6]
c) Explain how the Weiner filter is used in image restoration. [6]

OR

- Q6)** a) Explain any three Geometric transforms in image restoration. [6]
b) Explain estimating the degradation function for image restoration. [6]
c) Compare in detail between image enhancement and image restoration. [6]
- Q7)** a) What is a pattern in images? Explain the different pattern classes in object recognition. [6]
b) What are the methods of object recognition? Explain the recognition based on decision-theoretic methods. [6]
c) Explain any one application of image processing as object recognition in detail. [5]

OR

- Q8)** a) Explain any one algorithm of content-based image retrieval. [6]
b) Write a short note on the following structural methods. [6]
i) Matching shape numbers ii) String matching.
c) Explain in detail deep learning using CNN. [5]

